

Foundation (Jalapeno)

- Q1.** Which lines are parallel if they have the same m value in $y = mx + b$?
- (A) Lines with the same y -intercept
(B) Lines with the same slope
(C) Lines with the same x -intercept
(D) Lines with the same equation
(E) Lines that are perpendicular
- Q2.** What is the equation of a line parallel to $y = 2x - 3$ that passes through $(0, 5)$?
- (A) $y = 2x + 5$
(B) $y = 2x - 5$
(C) $y = x + 5$
(D) $y = x - 5$
(E) $y = -2x + 5$
- Q3.** If a stock's price increases by 2 each day, what is the equation of the line representing this trend if it starts at 20?
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- (A) $y = 2x + 20$
(B) $y = x + 20$
(C) $y = 2x - 20$
(D) $y = x - 20$
(E) $y = -2x + 20$
- Q4.** Two parallel lines have equations $y = 3x + 2$ and $y = 3x - 4$. What is the vertical distance between them?
- (A) 3
(B) 4
(C) 6
(D) 8
(E) 2
- Q5.** A savings account earns 5% interest per year. If a person starts with 1000, what is the equation of the line representing the balance after x years?
- (A) $y = 1.05x + 1000$
(B) $y = 0.05x + 1000$
(C) $y = 50x + 1000$
(D) $y = 5x + 1000$
(E) $y = 1000 - 0.05x$
- Q6.** Which of the following lines is parallel to $y = -2x + 1$?
- (A) $y = 2x + 1$
(B) $y = -2x - 1$
(C) $y = x + 1$
(D) $y = -x + 1$
(E) $y = 2x - 1$
- Q7.** A tax consultant charges a flat fee of 50 plus 10% of the client's tax refund. If x is the tax refund, what is the equation for the total cost y ?
- (A) $y = 0.1x + 50$
(B) $y = 0.1x - 50$
(C) $y = x + 50$
(D) $y = 10x + 50$
(E) $y = 10x - 50$
- Q8.** Two lines are parallel if their slopes are m_1 and m_2 and...
- (A) $m_1 = m_2$
(B) $m_1 = -m_2$
(C) $m_1 + m_2 = 0$
(D) $m_1 - m_2 = 0$
(E) $m_1 \cdot m_2 = 1$

Open Ended Questions (FRQ)

- Q9.** Show that the lines $y = 4x - 3$ and $y = 4x + 2$ are parallel by finding their slopes and y -intercepts.
- Q10.** A company's profit y (in thousands of dollars) is related to the amount spent on advertising x (in thousands of dollars) by the equation $y = 2x - 5$. Find the equation of a line parallel to this that represents a profit of 15,000 when no money is spent on advertising.

Chef's Tips

- Ensure students understand the concept of slope and its relation to parallel lines.
- Emphasize the importance of identifying the y -intercept in equations of parallel lines.
- Provide examples that illustrate real-world applications of parallel lines, particularly in the context of business and finance.